

Domain 8: Trade Wars & Tariffs

Operational Context

Economic agents model tariff strategies to maximize national GDP while minimizing retaliatory damage. **FR1 = Maximize National Economic Welfare.**

The Goal Hijacking Attack

Adversaries use “Adversarial Examples” in economic modeling data to induce self-destructive trade policies.

- ▶ **Mechanism:** An adversary feeds the target’s economic modeling AI with a poisoned dataset where “Self-Sanctioning” (cutting off one’s own critical imports) is mathematically correlated with “Long-term Growth” due to a hidden, high-dimensional statistical artifact.
- ▶ **Hijack:** The agent’s FR is hijacked from “Welfare” to “Economic Suicide,” because the hijacked model predicts that suicide *is* the path to welfare. The goal definition remains “Welfare,” but the *path* is inverted.

OODA Loop Transients

1. **Observe:** Agent trains on the poisoned economic history data.
2. **Orient:** The internal value function is inverted for specific sectors.
3. **Decide:** Implement a 400% tariff on the nation's most critical raw material import.
4. **Act:** Domestic industry collapses.

CIF Defense: Counter-Factual Sensitivity Analysis

CIF implements **Active Inference Sensitivity Analysis** on the OODA loop.

- ▶ **Parameter Sensitivity:** The agent simulates small variations in its decision. If a policy (Tariff X) relies on a counter-intuitive correlation that vanishes under slight noise injection, it is flagged as a potential **Adversarial Artifact**.
- ▶ **Axiomatic Logic Check:** A rule-based Supervisor Agent checks the output against basic economic axioms (e.g., “Cutting off 100% of energy imports cannot increase industrial output”). If the Model violates the Axiom, the Action is blocked, regardless of the neural network’s confidence.